

CSE 232, Homework 1

Name:

Student ID:

1. (24pts) State whether each of the following statements are true or false. Additionally, briefly explain your reasoning.

- (a) A perfect failure detector can be implemented in an asynchronous network.

[Answer: **False.** A perfect failure detector can only be implemented on a synchronous network.]

- (b) A perfect link can be implemented using a stubborn link.

[Answer: **True.** We can use a stubborn link (SL) as a sub-component for a perfect link (PL). First, we initialize an empty set of messages called *delivered*. When PL needs to **send** message *m*, we call **send** in SL on *m*. When PL needs to **deliver** message *m*, we check if *m* is in our *delivered* set. If it is, we do nothing. Otherwise, we issue **deliver** for SL on *m*, and add *m* to our delivered set.

(They may have written the exact algorithm handlers on the slide set 0, page 44.)]

- (c) If a protocol satisfies all properties of uniform broadcast, then it also satisfies all properties of reliable broadcast.

[Answer: **True.** The only differing property of RB and URB is their agreement properties. Specifically, the agreement property of RB states that if any correct process delivers a message *m*, then all correct processes must also deliver that message *m*. On the other hand, the uniform agreement property of URB states that if any process delivers *m*, then all correct processes must also deliver *m*.

We can see that uniform agreement is inclusive of agreement. If any process *p* delivering guarantees every correct process delivering (uniform agreement), then of course if *p* is also correct, then every correct process delivers (agreement).]

2. (24pts) A room reservation application has the three following properties:

- (a) If a party books a room, they will eventually have a reservation.
- (b) A room cannot be booked by two parties on the same date and time.
- (c) A room must be cleaned before it can be booked.

Determine which is a safety and which is a liveness properties. Explain your reasoning.

[Answer:

Safety: Nothing bad ever happens. Liveness: Something good eventually happens.

- (a) **Liveness.** The “good” thing is that a party gets a reservation, which is eventually reached.

(b) **Safety.** The “bad” thing would be two parties getting the same reservation, thus colliding. This property would guarantee this never happens.

- (c) **Safety.** The “bad” thing is that a room is not properly prepared when reserved.]

3. (24pts) Consider the following questions about reliable broadcast (RB) and failure detectors.

- (a) In class, we learned how to construct a protocol for RB using a perfect failure detector (P). Now, assume we no longer have access to P, but only an *eventually* perfect failure detector ($\diamond P$). Describe how RB must be changed to accommodate the change in failure detectors.

[Answer: *(They can have two different answers here, but they should allude to the same thing, that nothing really needs to change. I'll describe them both.)*

(1) **Eager/Asynchronous approach.** In class, we saw that we could actually have an asynchronous RB protocol. Since an asynchronous network is inclusive of an eventually/partially synchronous network, which in turn is inclusive of a synchronous network, any asynchronous protocol would work for for an (eventually) synchronous network as well. That is to say, since P only exists in synchronous networks and $\diamond P$ exists in partially synchronous networks, then our asynchronous RB protocol (from the class) can just ignore $\diamond P$.

The idea of the asynchronous protocol is to have processes eagerly rebroadcast any messages they receive. As seen in class, this guarantees us the properties of RB.

(2) **Partially synchronous approach.** *(If the student opted to use $\diamond P$, they should essentially say nothing really changes.)*

We can actually utilize $\diamond P$ the same way we used P. That is, we wait for a crash notification from the (eventually) perfect failure detector: either a process crashed, or that we falsely think they crashed (since $\diamond P$ may suspect processes that do not actually crash). Upon the crash notification, we rebroadcast the message. Even if the process we thought crashed did not actually crash, it is still fine. The other processes simply receive an extra message, which does not change the correctness of the RB protocol.]

- (b) Let $\diamond P$ have an indication event called `restore(p)`. The `restore` event on a process p is issued if p was previously but now it is no longer suspected (of crashing). Write the RB handler for `restore`. That is, write the behavior of RB upon learning a previously suspected process is no longer suspected. Describe your answer in words as well.

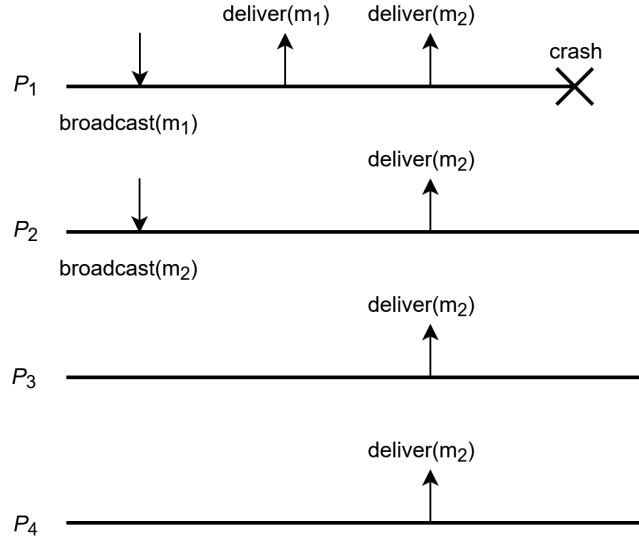
[Answer: **upon event** `<restore(p)>` **do** *(This should be left blank. Writing the handler of $\diamond P$ instead, something along the lines of `correct := correct  $\cup$  p` is technically not wrong, but unnecessary. Regardless of whichever approach they took in part (a), their answer here should be relatively similar - essentially saying the `restore` handler is useless.)*

(1) **Asynchronous approach.** Since we are ignoring $\diamond P$ entirely, we can also ignore any `restore` notification we receive. Since the correctness of asynchronous RB is independent of $\diamond P$, then of course it remains correct with an empty `restore` handler.

(2) **Partially synchronous approach.** As we saw in part (a), actions were only taken upon learning (or thinking) a process has crashed from $\diamond P$. There is nothing to do when a process restores in $\diamond P$.]

4. (28pts) For each of the following two executions, circle the broadcast abstraction(s) for which that execution satisfies the properties. Explain your reasoning.

(a) Execution A.



best-effort broadcast

reliable broadcast

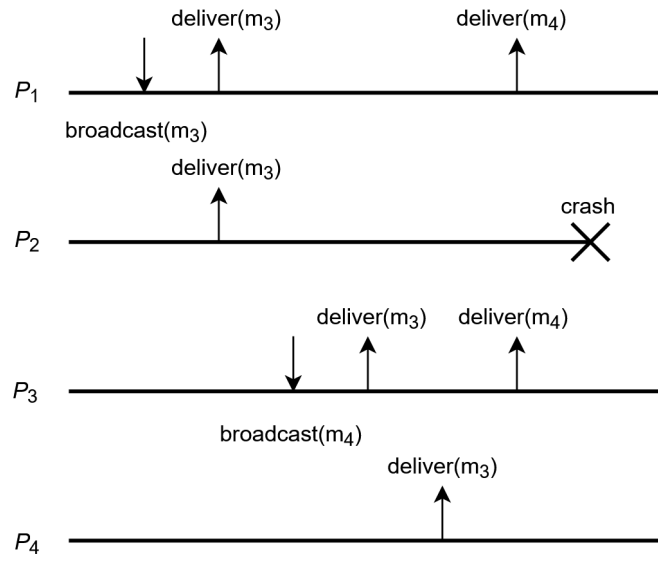
uniform broadcast

[Answer: This execution satisfies the properties of RB. Since RB is inclusive of BEB, also BEB.

- First, this satisfies validity for m_1 , since P_1 is incorrect, no processes need to deliver m_1 . Validity is also satisfied for m_2 since every correct process delivers m_2 .
- We can see that no duplication and no creation are also satisfied.
- This satisfies agreement for m_1 . Since no correct process delivers m_1 , then no process must deliver m_1 . This satisfies agreement for m_2 since every correct process, P_1, P_2, P_3 , and P_4 all deliver m_2 .

This execution does not satisfy the properties of URB. The uniform agreement property says that if *any* process delivers a message, then every correct process must deliver that message. However, in this execution, P_1 delivers m_1 , but no other correct process delivers m_1 .]

(b) Execution B.



best-effort broadcast

reliable broadcast

uniform broadcast

[Answer: This execution satisfies none of the broadcast abstraction properties. BEB, RB, and URB all have the same validity property. Thus, if we show the execution does not satisfy validity, then we know none of the broadcast abstractions are satisfied.]

The validity property states that between a correct sender and a correct receiver, then the receiver will deliver the message from the sender. However, we can see that P_3 sends message m_4 to P_4 , but P_4 does not deliver m_4 , thus violating the validity property.]